

Universal Digital Agents That Learn, Remember, and Act

Research Internship / Talent Program Interview

GUI Agents

Continual Learning

Real Systems

From research questions to working systems and measurable improvement loops

Expected Jun 2028 | Zhejiang University APRIL Lab



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Candidate Snapshot | Research Focus and Impact

Research vision: universal agents that continuously learn, remember, and act in real digital environments

Mobile GUI agents, long-horizon tasks, memory evaluation, and reinforcement-learning post-training

5

First-author works

TMLR / ACL / arXiv

415

Google Scholar citations

Academic impact

122

First-author citations

Core research

632

GitHub stars on first-author projects

Open-source impact

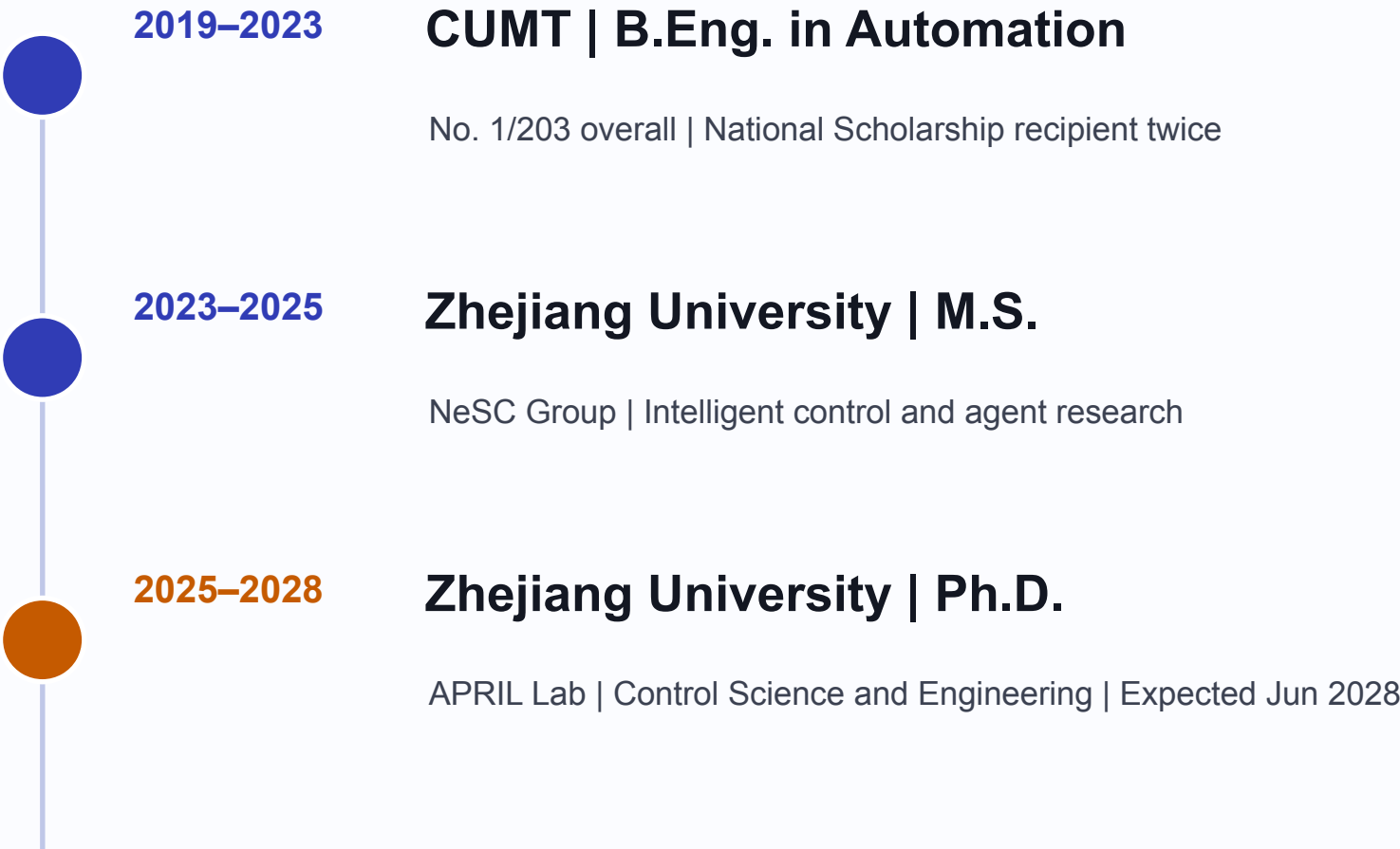
Jun 2028

Expected graduation

Zhejiang University Ph.D.

Source: CV and personal homepage; metrics as of 2026-08-09

Education and Growth



A continuous agent-research trajectory

Source: CV and personal homepage; metrics as of 2026-08-09

Industry Research | From Prototypes to Real Systems

Aug 2024–Aug 2025

vivo AI Lab

- PhoneGPT and demonstration learning
- Online evaluation and memory benchmarking
- Connecting research questions to real phone scenarios

Aug 2025–Jul 2026

Kuaishou

- Agent post-training and reinforcement learning
- Long-horizon context management
- App defect detection and automated evaluation

Research | Testable questions

Engineering | Data and eval loops

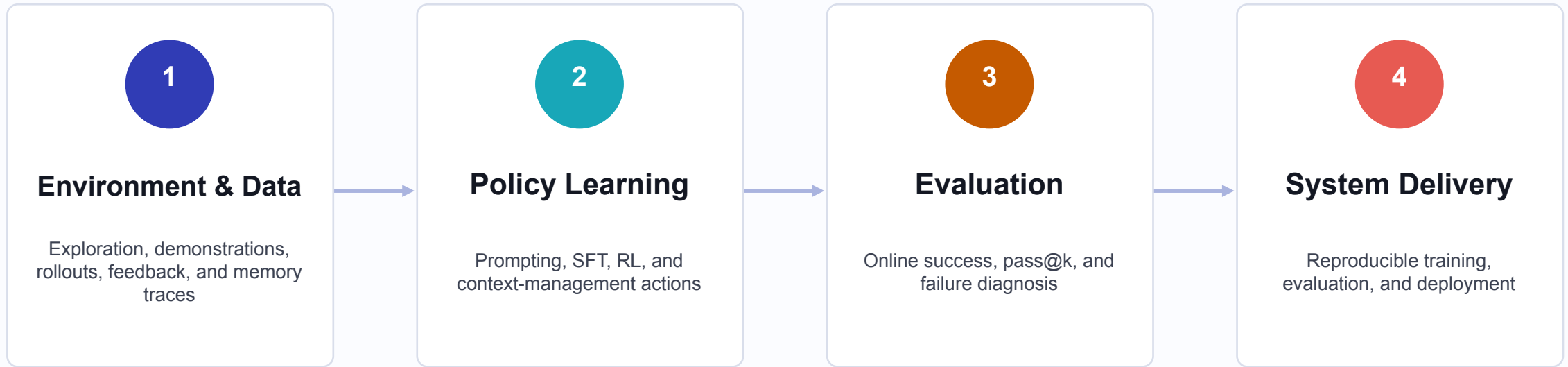
Application | Product workflows

Strength: defining the research question and carrying it through environment, data, training, and evaluation

Research Thesis | Continuous Improvement in Real Environments

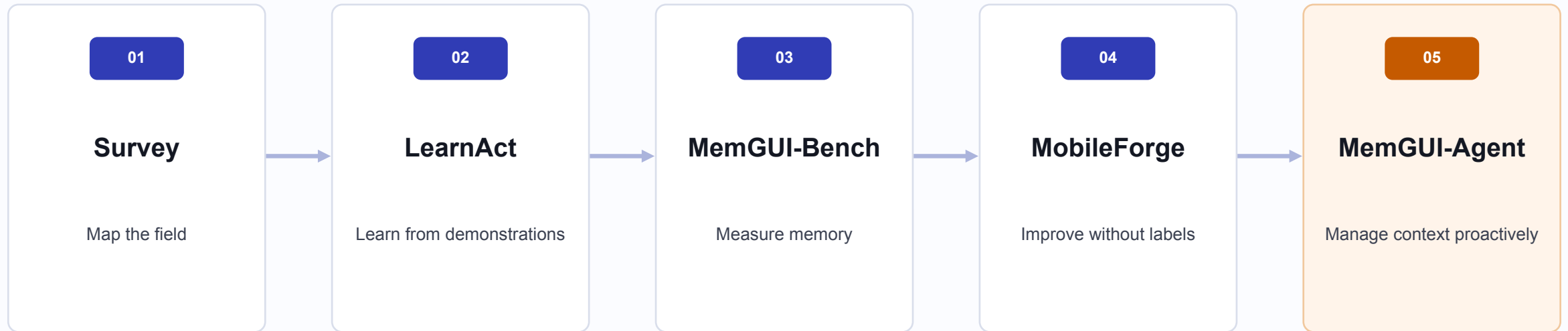


End-to-End Ownership | From Question to System



Find the bottleneck → make it measurable → close the loop → test transfer

Five First-Author Works, One Coherent Research Arc



Field map → demonstration learning → memory evaluation → self-improvement → long-horizon agents

Next: problem, method, evidence, and research value for each paper

LLM-Powered GUI Agents in Phone Automation: Surveying Progress and Prospects

A systematic map of frameworks, models, data, evaluation, and future directions for phone GUI agents

CORE QUESTION A rapidly growing field lacked a shared vocabulary connecting academic progress with practical agent design.

TMLR 2025

Survey & Taxonomy

First Author

arXiv:2504.19838

Technical Evolution | From Fixed Scripts to Adaptive Agents

Traditional automation exposes three fundamental gaps under dynamic interfaces, long-tail tasks, and natural-language intent

Traditional automation

Fixed paths and templates

Predefined scripts, selectors, and rules are brittle under UI updates, expensive to maintain, and difficult to transfer across apps.

Multimodal perception

Screenshot / UI Tree / OCR

Agents combine screenshots, UI trees, OCR, and Set-of-Marks, moving from coordinate matching to semantic interface understanding.

Agent brain

Planning, reasoning, reflection

Single-agent, multi-agent, and plan-then-act designs trade execution efficiency against modularity and complex-task decomposition.

Action interface

Touch, tools, and APIs

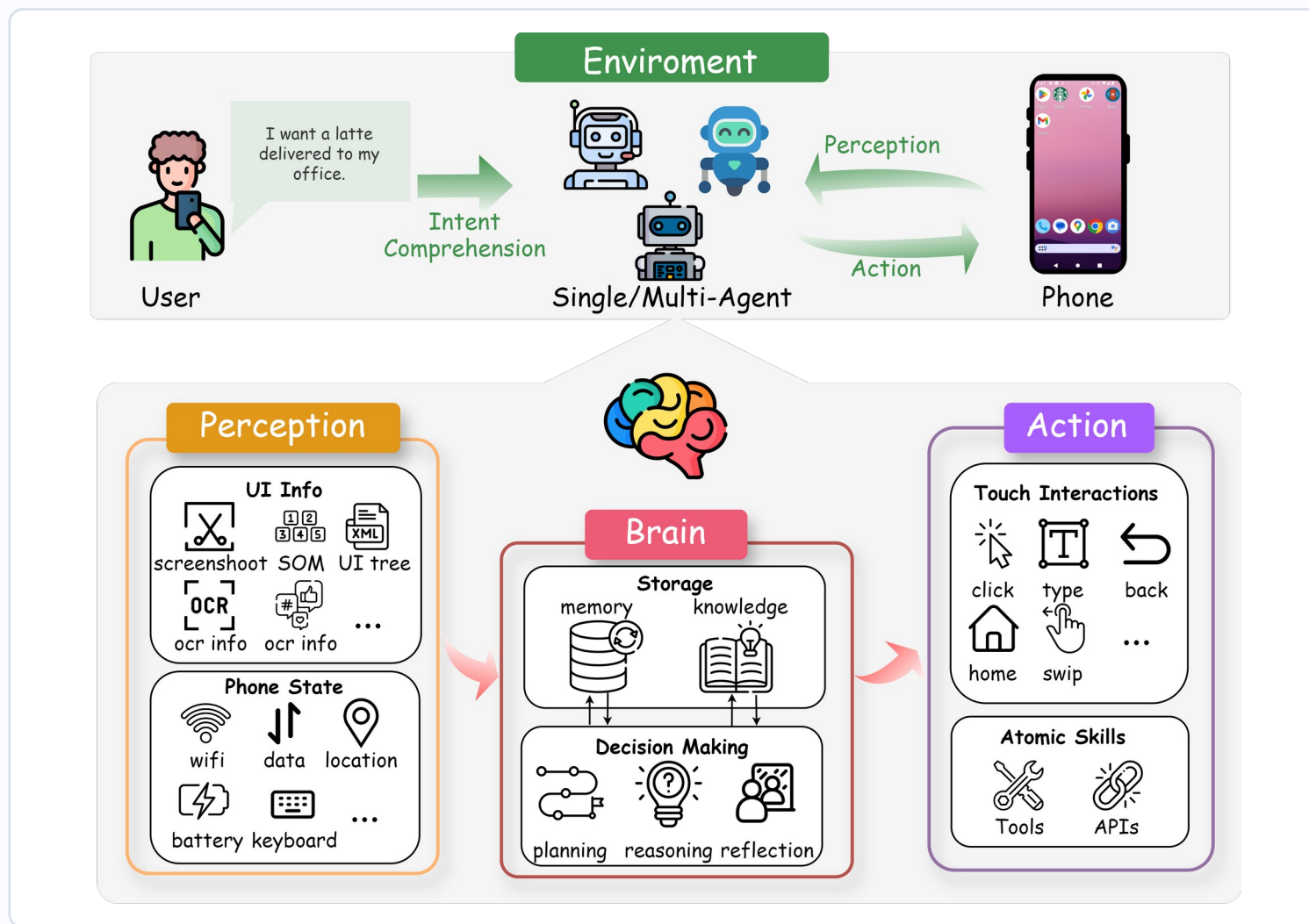
Click, type, and swipe are extending toward tools and APIs, but dynamic content still demands more expressive action spaces.

ORIGINAL PAIN POINTS

Limited generality, high maintenance overhead, and weak intent understanding drive the shift from scripts to LLM agents.

Source: Liu et al., arXiv:2504.19838; TMLR 2025

Unified Framework | From User Intent to GUI Action



A Three-Layer Research Map

- Agent frameworks: single-agent, multi-agent, and plan-then-act
- Modeling: prompt engineering versus training-based methods
- Resources: datasets, benchmarks, metrics, and automated evaluation
- Core loop: perceive the GUI, understand intent, decide, and act

Turn fragmented progress into reusable agent-design guidance

Research Roadmap | Toward Deployable Phone Agents

The shift from static scripts to adaptive agents exposes bottlenecks beyond model capability

Data

Coverage diversity

Long-tail apps and real tasks

On-device

Deployment efficiency

Compute, latency, and energy

Users

Personal adaptation

Preferences, history, and feedback

Safety

Reliability and privacy

Sensitive actions and risk control

RESEARCH VALUE A common vocabulary and evaluation lens that anchors later work on demonstration learning, memory, and continual improvement.

A clear field map reveals the system bottlenecks worth solving

System Trade-offs | The Frontier Is More Than Scaling Models

The survey reframes future work as a set of coupled system trade-offs

Framework design

Speed–memory–cost trilemma

Richer perception and multi-agent systems raise capability but also latency, context, and API cost; resources should scale with task complexity.

Modeling path

Generality–specialization gap

Prompting adapts easily but inherits the base-model ceiling; SFT/RL is stronger but data-intensive, motivating hybrid training.

Data and evaluation

Offline–online capability gap

Strong single-step grounding does not imply multi-step task success; benchmarks need cross-app composition, efficiency, and robustness.

On-device deployment

Size–latency–performance trilemma

Compact models improve latency and privacy, but must remain robust to UI changes, interruptions, and variable networks.

TRUSTWORTHINESS

Environmental injection, data poisoning, privacy leakage, and model bias require end-to-end defenses, monitoring, and privacy-preserving training.

LearnAct: Few-Shot Mobile GUI Agent with a Unified Demonstration Benchmark

Improving long-tail mobile-task generalization from a few visual demonstrations

CORE QUESTION Agents struggle to generalize to long-tail tasks, while the field lacked a unified benchmark for few-shot demonstration learning.

ACL 2026

2,353 Tasks

6 Backbones

arXiv:2504.13805

LearnGUI | Controlled Evaluation of Few-Shot Demonstration Learning

The benchmark covers offline step execution and online end-to-end interaction while controlling demo–query similarity

LearnGUI-Offline

2,252 tasks

Reorganizes AMEX into demonstration–query combinations and evaluates step type, step completion, episode type, and episode completion.

LearnGUI-Online

101 tasks

Adds high-quality human demonstrations to AndroidWorld and measures end-to-end success under real interaction.

Overall scale

2,353 instructions / 73 apps

Tasks average 13.2 steps and share a unified action space for both general-purpose and GUI-specialized agents.

Controlled variables

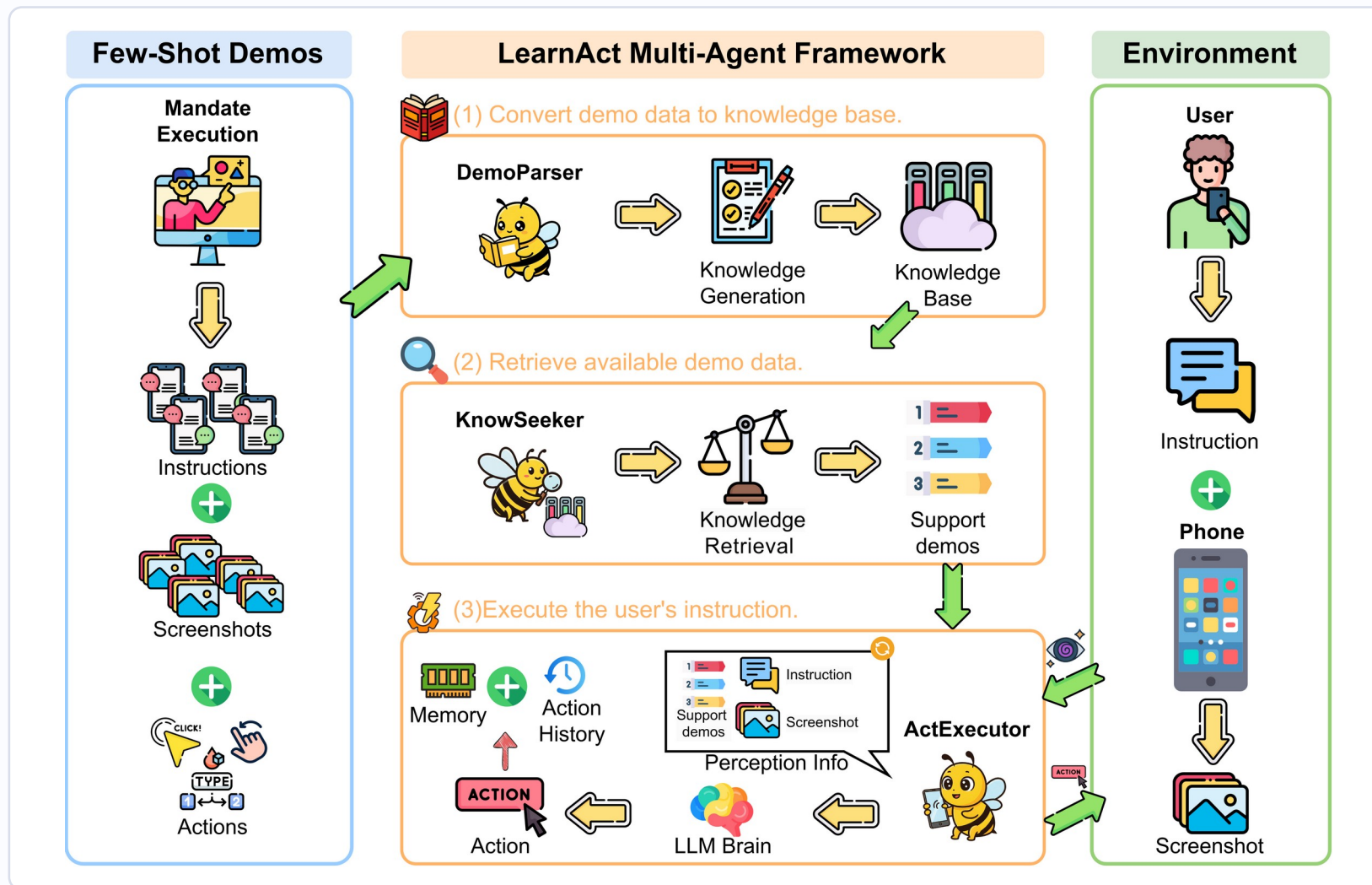
1 / 2 / 3-shot

Systematically varies instruction, UI, and action similarity to separate quantity effects from relevance effects.

UNIFIED ACTION SPACE

CLICK, TYPE, SWIPE, BACK, HOME, ENTER, and TASK_COMPLETE align offline and online evaluation.

LearnAct | Turning Visual Demonstrations into Retrievable Knowledge



Three Modules

- DemoParser extracts structured knowledge from demonstrations
- KnowSeeker retrieves support demos by instruction, UI, and action similarity
- ActExecutor combines demos, screenshots, history, and memory for execution

Do not copy trajectories; extract, retrieve, and reuse demonstration knowledge

Results | Consistent Gains Across Model Families

LearnGUI: 2,252 offline tasks + 101 online tasks across 73 applications

38.5→58.9%

Gemini-2.5-Pro offline

+20.4 percentage points

18.1→32.8%

UI-TARS-7B-SFT online

+14.7 pp; +81.2% relative

6

Backbone models

General-purpose and GUI-specialized

KEY FINDING

Demonstration learning benefits both general-purpose and GUI-specialized models, providing a low-cost interface for long-tail adaptation.

A few high-quality demonstrations can materially improve online generalization

Component Ablation | Generation and Retrieval Are Complementary

Average accuracy across nine apps shows that each component helps independently and their combination is strongest

Baseline

19.3%

Without DemoParser or KnowSeeker, the agent relies only on the current instruction and visual state.

KnowSeeker only

40.6%

Relevant-demo retrieval helps substantially, but raw trajectories lack semantic action descriptions and explicit memory.

DemoParser only

41.6%

Structured knowledge improves reuse, but without task-aware retrieval irrelevant demonstrations can enter the context.

Full LearnAct

51.7%

Knowledge generation and top-k semantic retrieval combine to deliver the best average over nine applications.

K-SHOT EFFECT

Gemini-Pro-1.5 step-complete accuracy rises from 19.3% to 57.7% at 3-shot; demos improve complete execution more than action-type recognition.

MemGUI-Bench: Benchmarking Memory of Mobile GUI Agents in Dynamic Environments

A systematic benchmark for short-term memory, long-term learning, and execution efficiency

CORE QUESTION Existing benchmarks contain only 5.2–11.8% memory-related tasks and cannot evaluate cross-session learning.

128 Tasks

26 Apps

11 Agents

arXiv:2602.06075

Tasks and Metrics | Measuring Short-Term Memory and Long-Term Learning

Task design, snapshot environments, repeated attempts, and memory-specific metrics form one evaluation loop

Task scale

128 tasks / 26 apps

Covers 68 real scenarios, averages about 36.2 golden steps, and supports information transfer across as many as four apps.

Memory density

89.8% memory-intensive

Tasks require retaining intermediate results across time or moving and reusing information across screens and apps.

Short-term memory

SR / IRR / MTPR

Success Rate measures completion, IRR measures correct information retention, and MTPR isolates memory-specific proficiency.

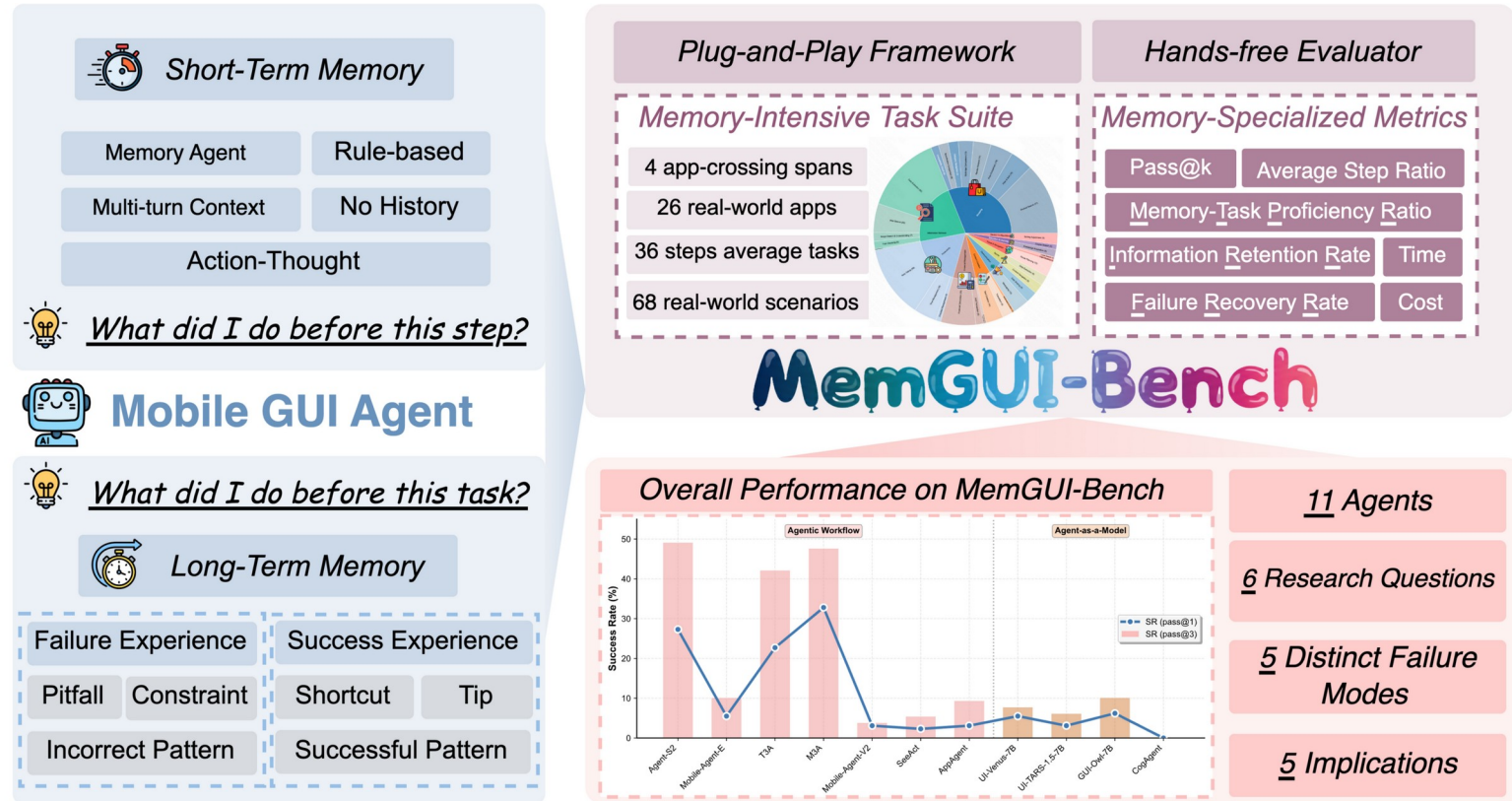
Learning and efficiency

pass@k / FRR / cost

Repeated attempts measure learning from failure while step ratio, time per step, and cost per step capture deployability.

MIRROR TASK PAIRS The 128 tasks form 64 mirror pairs with similar cognitive demands but different requirements, separating answer memorization from experience transfer.

MemGUI-Bench | Taxonomy, Tasks, and Evaluation in One System



Evaluation Design

- Taxonomy: short-term retention and long-term cross-session learning
- Tasks: 128 tasks across 26 apps; 89.8% require cross-temporal or cross-spatial memory
- Protocol: pass@1 / pass@k for repeated attempts and learning
- MemGUI-Eval: three-stage Progressive Scrutiny and seven hierarchical metrics

Findings | Standard Benchmarks Hide the Real Memory Gap

Six research questions across eleven state-of-the-art agents

4–10×

Memory capability gap

Standard vs memory-intensive tasks

−16–40 pp

Cross-app degradation

Primary memory bottleneck

+18.8 pp

Long-context gain

RQ4

+21.9 pp

Long-term memory gain

RQ5

FAILURE DIAGNOSIS Five failure modes: partial, process, and output memory hallucination, knowledge deficiency, and intent misunderstanding, leading to five design implications.

A trustworthy benchmark is prerequisite to judging whether memory actually works

Progressive Scrutiny | Cost-Efficient and Interpretable Evaluation

The evaluator increases compute only when evidence is insufficient, avoiding expensive full visual reasoning for every long trajectory

Stage 1

Cost-effective triage

Inspect only terminal screenshots and raw action logs; declare success only when requirements are unambiguous, otherwise escalate.

Stage 2

Full semantic analysis

Generate before–after descriptions for every step, then let a semantic judge combine instruction, actions, and state changes.

Stage 3

Targeted visual verification

Retrieve decisive screenshots for unresolved conditions such as products, values, settings, or cross-app information.

Failure taxonomy

Five memory failures

Partial, process, and output memory hallucination, knowledge deficiency, and intent misunderstanding localize the real bottleneck.

DEPLOYMENT INSIGHT

Short-term memory is mandatory, long-term memory is beneficial but optional, and token budgets constrain deployment more than step budgets.

MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization

Explore target apps, mine curricula, evaluate rollouts, and optimize policies without manual annotations

CORE QUESTION Target apps are numerous and fast-changing, making adaptation with human tasks, demonstrations, and reward labels unscalable.

MobileGym

HiFPO

RL Post-Training

arXiv:2606.19930

MobileGym | Building the Annotation-Free Adaptation Substrate

Annotation-free does not mean supervision-free; target-app interaction is automatically converted into tasks, rollouts, and hierarchical feedback

Target-app exploration

Function-aware DFS

App anchors and screenshots guide goals; depth-first traversal records reachable screens, controls, actions, and transitions into evidence pool Z.

Curriculum mining

MobileGym-Curriculum

Checks trajectory coherence and completion, then generates executable, verifiable tasks with diverse difficulty.

Multi-attempt rollout

K attempts + prior hints

The policy retries the same task, with later attempts receiving corrective hints distilled from prior failures.

Hierarchical evaluation

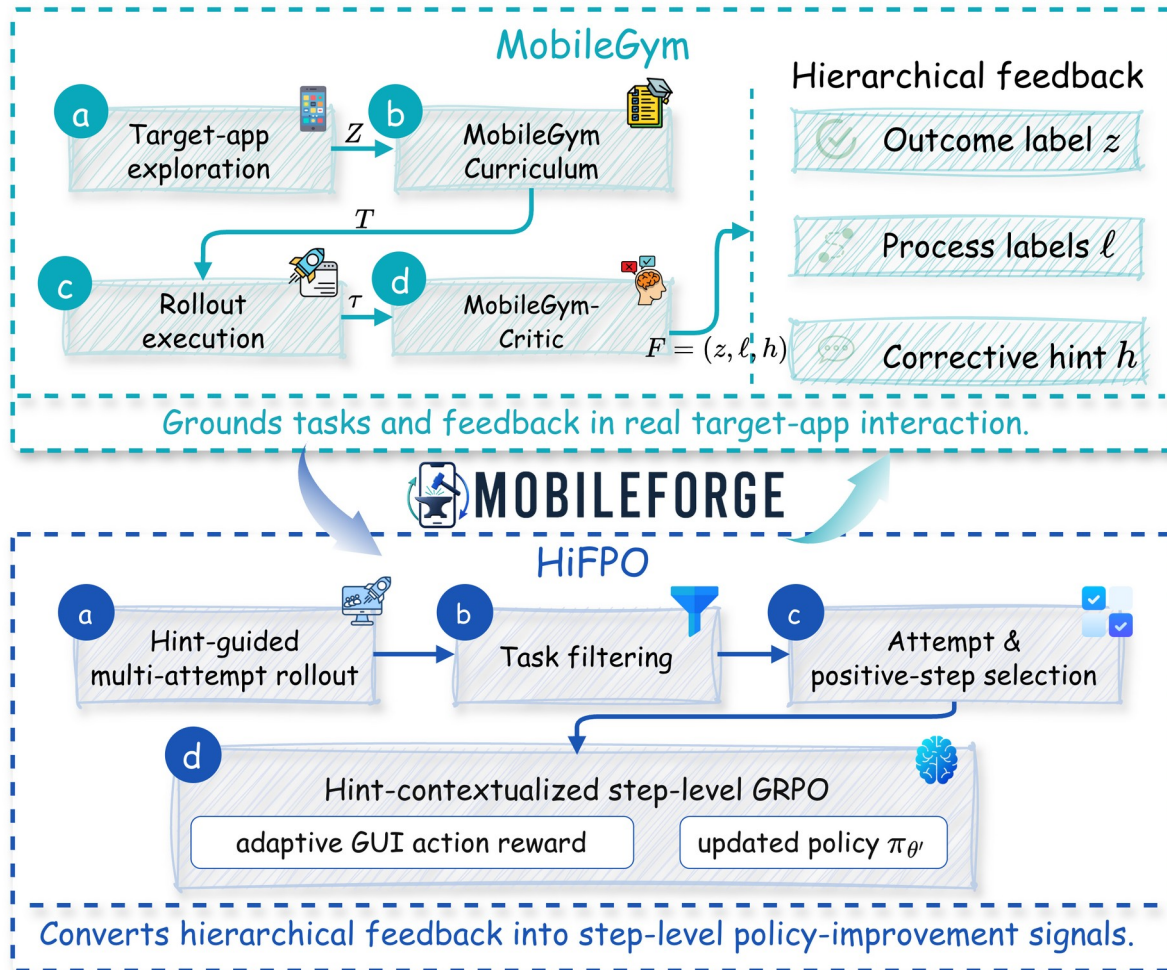
$z + l + h$

MobileGym-Critic outputs trajectory outcome z, step-level process labels l, and corrective hint h instead of one coarse reward.

CLOSED LOOP

$\text{Explore}(E) \rightarrow \text{Curriculum}(Z) \rightarrow \text{Rollout}(\pi, x, \eta) \rightarrow \text{Critic}(x, \tau) \rightarrow \text{HiFPO}(\theta, T, \tau, F).$

MobileForge | A Real-Interaction Self-Improvement Loop



Two Coupled Components

- MobileGym: target-app exploration, curriculum mining, rollout execution, and hierarchical evaluation
- Hierarchical feedback: outcome labels, process labels, and corrective hints
- HiFPO: multi-attempt rollouts, task filtering, and positive-step selection
- Hint-contextualized step-level GRPO converts feedback into learnable signals

The environment generates data; hierarchical feedback becomes policy-improvement signal

Results | Automatically Generated Data Can Produce Strong Policies

Validated on AndroidWorld and MobileWorld using only automatically generated adaptation data

67.2%

Qwen3-VL-8B Pass@3

AndroidWorld; near the closed-data base at 69.0%

77.6%

ForgeOwl-8B Pass@3

AndroidWorld

41.0%

ForgeOwl-8B success

MobileWorld GUI-only

KEY RESULT

MobileForge improves both generalist and GUI-specialized models; ForgeOwl-8B is the strongest open-data mobile GUI agent in the paper.

Turn real target apps into a continuously improving agent factory

HiFPO Ablations | What Actually Drives Improvement

Corrective hints, task filtering, and hint-contextualized GRPO address exploration, data quality, and optimization respectively

Corrective hints

52.0% → 77.0%

On 200 generated tasks, hints raise overall success by 25.0 pp and Pass@3 by 23.5 pp across repeated attempts.

Task filtering

Keep mixed + hard

Remove mastered all-success tasks while retaining mixed and all-fail tasks; 1,910 samples from 167 tasks yield 15/117 on MobileWorld.

Training objective

50.9% Pass@1

With 900 tasks, hint-contextualized GRPO reaches 59/116 on AndroidWorld versus 40.5% base and 44.0% no-hint SFT.

Step selection

Attempt + positive step

Filter unlearnable or mastered tasks, then construct hint-conditioned step-level GRPO updates from successful attempts and reasonable steps.

KEY LESSON

More synthetic data is not automatically better: hints improve exploration, filtering improves data value, and step-level feedback improves credit assignment.

MemGUI-Agent: An End-to-End Long-Horizon Mobile GUI Agent with Proactive Context Management

Making context management a first-class action inside an end-to-end policy

CORE QUESTION ReAct passively accumulates history, causing prompt growth, noise, and dilution of critical cross-app facts on long-horizon tasks.

ConAct

MemGUI-3K

8B SFT

arXiv:2606.19926

MemGUI-3K | Teaching Smaller Models to Control Context

ConAct is more than an output format: smaller models must learn when to fold, when to write memory, and how to describe a step

Motivation experiment

The protocol alone is insufficient

Zero-shot ConAct helps only Qwen3-VL-235B-Thinking; smaller models and 235B-Instruct often regress.

Teacher rollouts

235B-Thinking + ConAct

Tasks run in the MemGUI-Bench snapshot environment with budget 2.5g+1 and are judged by Progressive Scrutiny.

Trajectory filtering

reasonable / unreasonable

Recovery loops, repeated errors, and counterproductive actions are removed while successful context-control behavior is retained.

Training scale

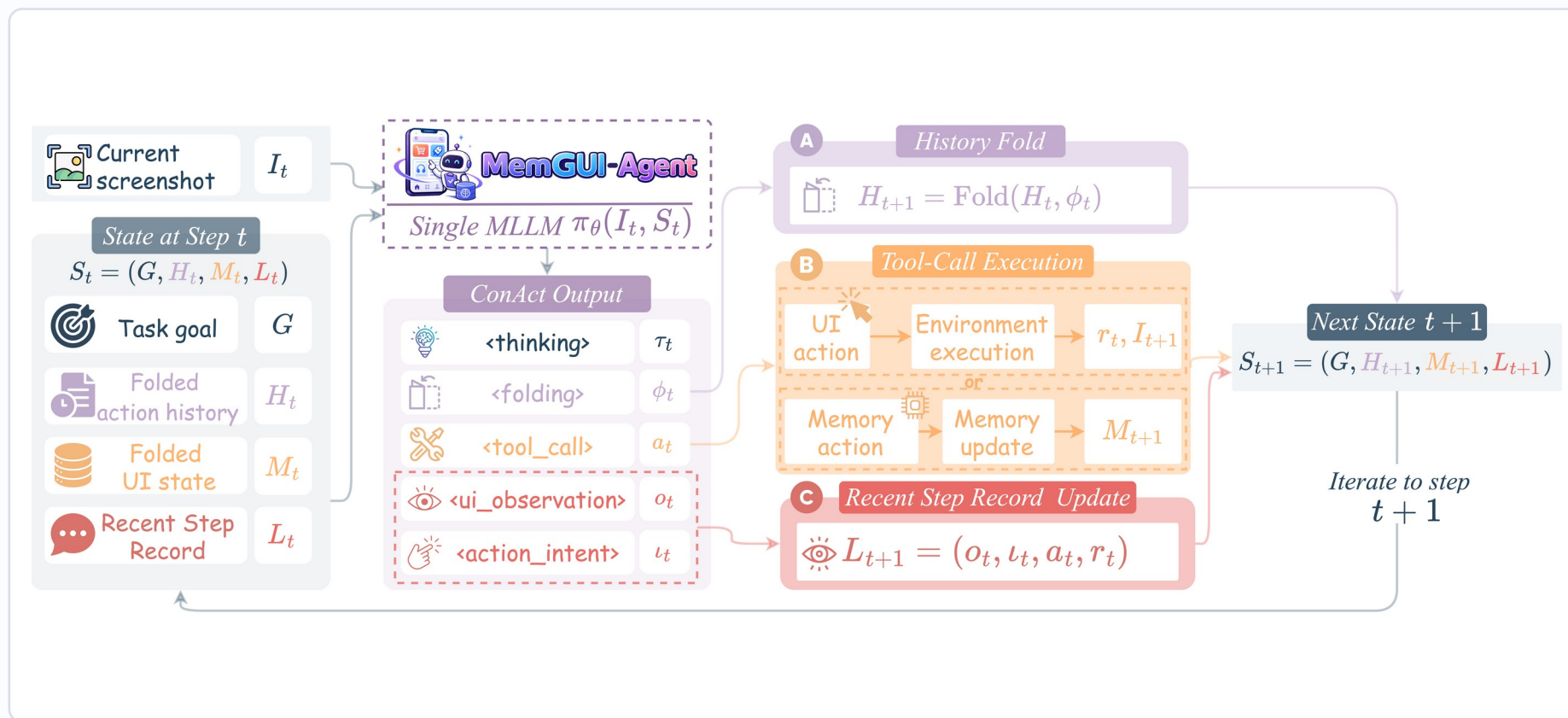
2,956 trajectories / 64,430 samples

Covers 26 apps, splits into 57,951/6,479 train/test steps, and has zero overlap with 128 MemGUI-Bench evaluation tasks.

TRAINING

LoRA SFT starts from Qwen3-VL-8B-Instruct; each sample contains structured context, screenshot, and the full five-part ConAct response.

ConAct | Context Management as Action



One Policy, Joint Decisions

- Folded action history compresses completed sub-tasks
- Folded UI state preserves critical interface facts
- Recent step record retains immediate observations and intent
- One forward pass emits reasoning, folding, tool call, observation, and next intent

While acting, the agent decides what to preserve and what to compress

Results | Lower Context Cost, Stronger Long-Horizon Performance

Zero-shot ConAct with 235B and supervised MemGUI-8B-SFT

~1.5k

Input tokens saved by step 150

Compared with ReAct

62.5%

235B MemGUI-Bench Pass@3

Pass@1 = 37.5%

2,956

MemGUI-3K trajectories

64,430 SFT samples

17.9%

8B MobileWorld success

+8.5 pp over backbone

TRANSFER

Zero-shot 235B ConAct sets a new MemGUI-Bench SOTA; training on MemGUI-3K gives the best open-data 8B result and transfers to MobileWorld.

A long-horizon agent must learn to manage its own working context while acting

Component Ablation and Failure Analysis | Three Complementary Mechanisms

History folding controls growth, UI memory preserves exact facts, and self-description provides reusable semantics for both

UI memory only

Pass@1: 5.0% → 17.5%

Persistent state is valuable, but without folding and step semantics long-horizon context still accumulates noise.

History folding only

22.5% / 32.5%

Pass@1 improves but Pass@3 remains limited, showing that compression alone cannot preserve critical facts reliably.

Self-description only

Pass@1: 25.0%

Explicit UI observation and action intent improve reuse, but persistent memory and span compression are still absent.

Full ConAct

40.0% / 62.5%

Pass@1/Pass@3 exceed every single component; failures fall from 99 to 58, with process -42% and output -57%.

OFFLINE SKILLS

8B SFT: action match 29.2%→36.3%, memory trigger F1 19.9%→48.0%, deep-fold ratio 8.8%→26.1%, and format compliance 99.9%.